QC STORY PROJECT

PROJECT TITLE :- To eliminate concern less torque of hand brake mtg. & console mtg. at trim 2 in TCF

Mahindra and Mahindra Ltd, Plant I, Scorpio Body shop.

	Plan	Actual
Start Date	**01/06/12**	**07/06/12**
Completion Date	**31/10/12**	**26/11/12**

Current Status:

completed

Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	Team Leader	Umesh kankal	Asst. Manger	Scorpio body shop	9604267879	Kankal.umesh@mahindra.com
2	Mentor	Suryavanshi. R.G	L4-DH	Scorpio body shop	9881378988	SURYAVANSHI.RAVINDRA@mahindra.com
3	Member 1	Vivek bhat	Dept. Manger	Scorpio body shop	9922438403	Bhat.vivek@mahindra.com
4	Member 2	Mangesh Gangurde	MQA- Asst. Manger	Scorpio body shop	9922644399	Gangurde.mangesh@mahindra.com
5	Member 3	Giri Debabrata	L8-O	SQA	9767744812	GIRI.DEBABRATA@mahindra.com

1. Select the Theme

Brief description of Problem

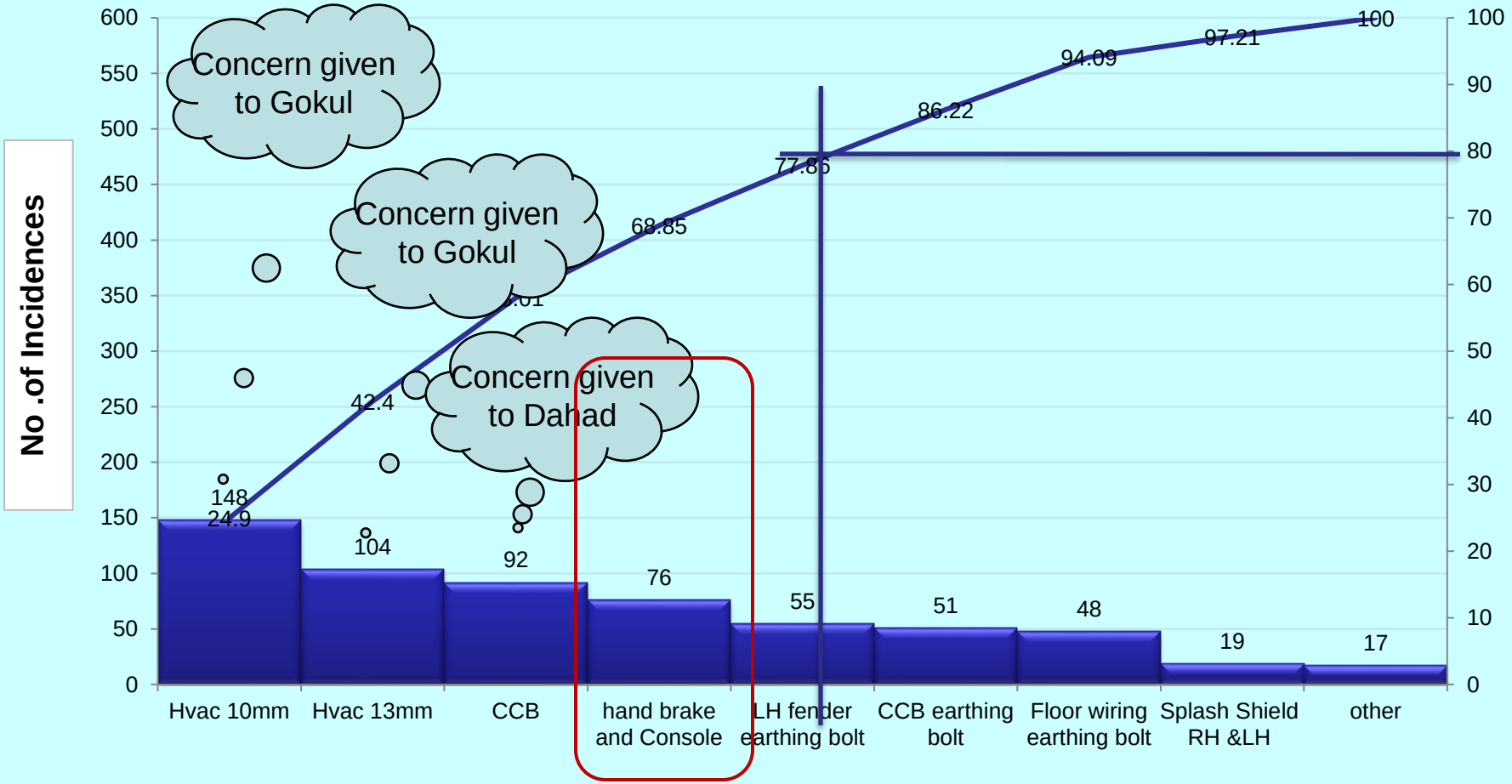
- **What:** Half torque for hand brake assly and console assly
- **Where:** At trim 2
- **When :** Daily(1 & 2 shift)
- **Who :** Stage operator/checkman/Final buyoff/CAI
- **How:** Rework qty. –36 nos. in the month of May 2012,
  40 nos. in June-12,
- **Rework time**: 20 min./ vehicle.

less torque hand brake and console mtg assly.

Avg(may & june) 38 BIW reworked for this quality concern.
Cost is -1.001 lacs Annualy

2. Justify the choice (reasons)

Pareto for Trim 2 Half Torque concern (2 Month data)



lessTorque Concern Reported at Trim 1

2. Justify the choice (reasons)

Sr No	Month	Problem Name	Location	Qty	Severity/Score	Interval (Freq)	Rework time(Cost)	Total
1	May 12 to June12	Less torque- H-VAC 10 mm	TCF	148	3	9	3	81
2	May 12 to June12	Less torque- H-VAC 13 mm	TCF	104	3	9	3	81
3	May 12 to June12	Less torque- CCB mounting	TCF	92	3	9	3	81
4	May 12 to June12	Less torque- console and hand brake mounting	TCF	76	9	9	3	243
5	May 12 to June12	Less torque- LH fender earthing bolt	TCF	55	3	9	3	81

Concern given to Gokul

Concern given to Gokul

Concern given toDahad

Due to High Severity, Occurrence and Rework time we have selected the "less torque- hand brake and console mtg bolt" issue for resolution.

2.b Immediate Containment Action

1) 100% bkt. position inspected at supplier end to ensure twisting of bkt.
2) A check point added in IQ checklist.
3) In TCF hand brake mounting monitoring started for every vehicle.

IQ-CHECK LIST -SCORPIO BODY SHOP

Sr No	BWT No	BWT Name	Phenomena	Physical Phenomena	29.10.2012					
1	BWT 1	Underbody (Plenum,Dash,Front end to Rear floor) Z-02,Z-14,Z-03,Z-05,Z-06	Mismatch	Hole mismatch accelerator mtg. bkt						
			Mismatch	Hand brake and console mtg						
			Damage	Head lamp mounting and Radiator mtg bkt.(SC 84)						
			Bent 14.06.12	Flasher mounting stud						
			Mismatch	Front seat mounting channel hole			X			On boundry
2	BWT 2	Framing line(Balloning to Deburring) Z-08+Z22	Stud bend	WWB bracket						
			Burr removal	LH/RH A pillar,Spot on edge -D pillar closure				X		D pillar rh corner spot on edge
			Damage	ECU mtg bracket						
			Damage	Roof LH/RH						
3	BWT 3	Side RH Z12	Dent/Damage	A,B, C and D pillar						
			Spot weak	A pillar drip rail						
			Mismatch	D pillar flange						
			Damage	Rocker panel						
			Mismatch/Spatter	B and C pilar seat belt nut						
4	BWT 4	Side LH Z-12	Dent/Damage	A,B, C and D pillar						
			Spot weak	A pillar drip rail						
			Excess sealer	Fuel lid housing NOVA C V1 28.02						
			Damage	Rocker panel						
			Mismatch/Spatter	B and C pilar seat belt nut						
Audited by:-										
BWT Leader:-										
			Mismatch	LH frt door speaker mtg hole						

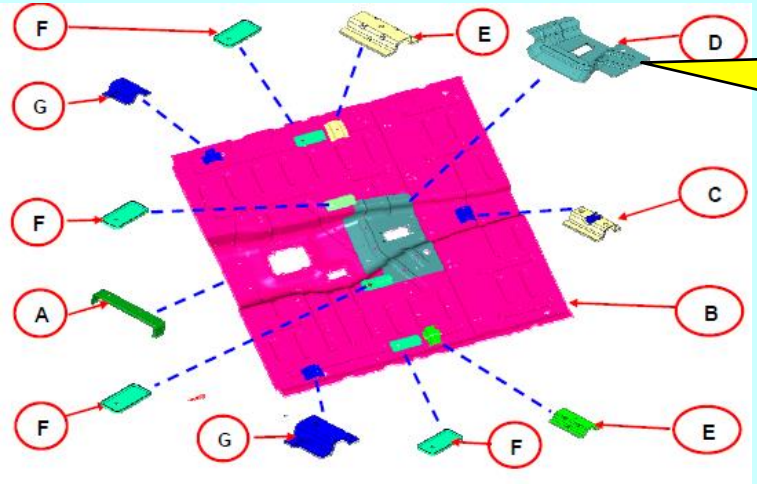
3. Understand the current situation

Process Flow

Front floor stn 10

Front floor stn 20

Front floor stn 25

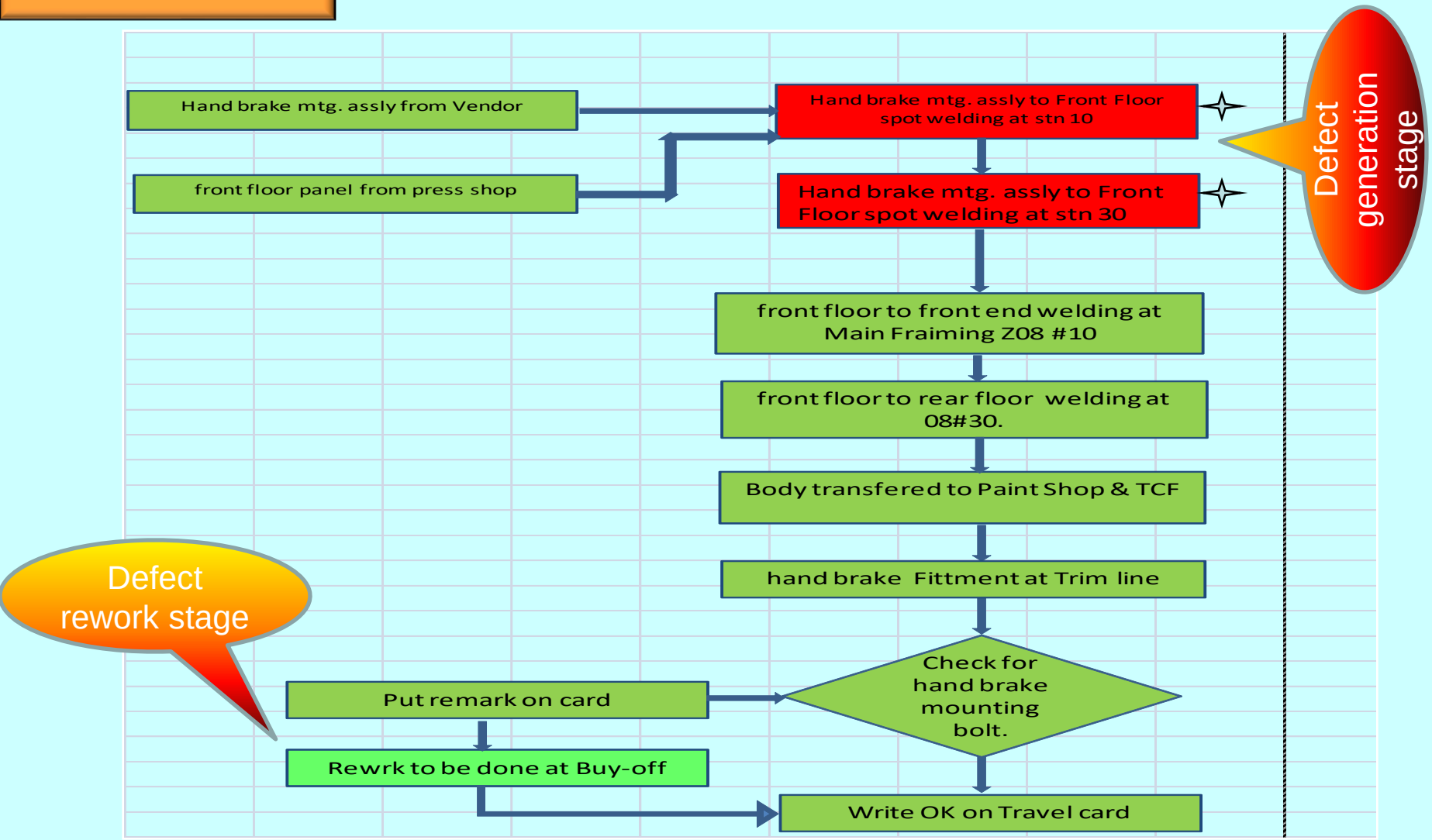


Hand brake and console mtg bkt to be fitted On front floor

Front floor stn 30

3. Understand the current situation

Flow chart



3. Understand the current situation

Observation of process

1) Console mounting bracket locating pin not inserting fully.

2) At front floor stn 30 spot on hand brake mounting bracket nut (possibility).

3. Understand the current situation

Observation of process

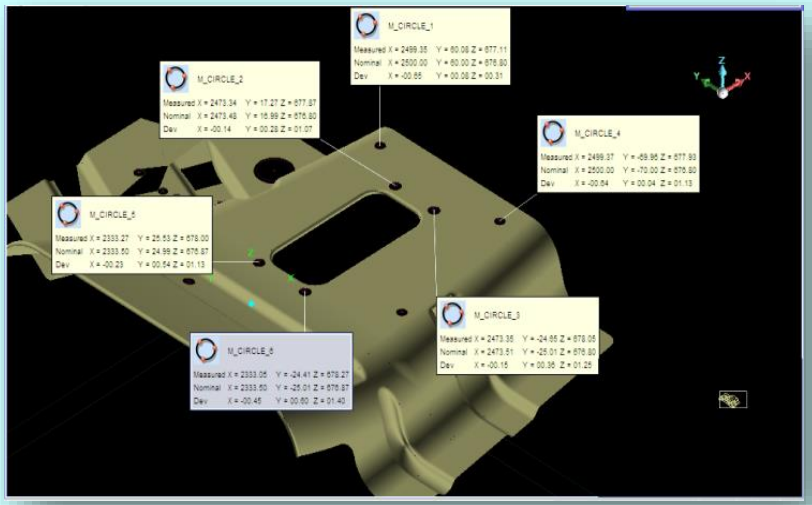
3) Hand brake mounting bracket held by clamp during spot welding operation.

4) Hand brake and console mtg brkt mismatch observed at front floor stn 10.

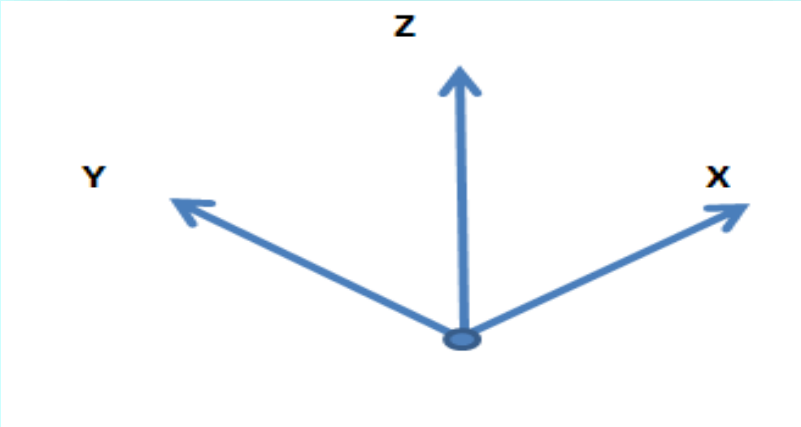
3. Understand the current situation

Observation of process

5) Variation found in hand brake mounting bracket wrt Front floor in X direction.



6) No shim ability found in X direction for correcting the position.



3. Understand the current situation

Observation of process

Vendor :-

1) At vendor end M/S MUSCO hand brake mounting nut plate spot welding with panel done on pedestal machine.

2) While spot welding operator using handy fixture .

3) No positive location found on handy fixture while spot welding.

4) Operator to operator variation found while spot welding.

5) Nut plate in tilt condition at supplier end.

TCF :-

1) hand brake bolt fitment done at Trim 2 line.

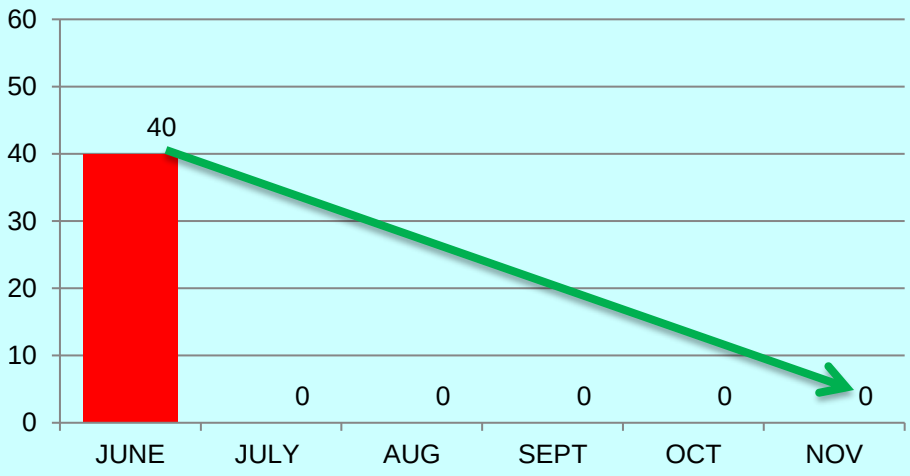
4. Select Target

Present Level:

Total 40 vehicles reported for the defect in june-12 month

Target:

Zero Defect



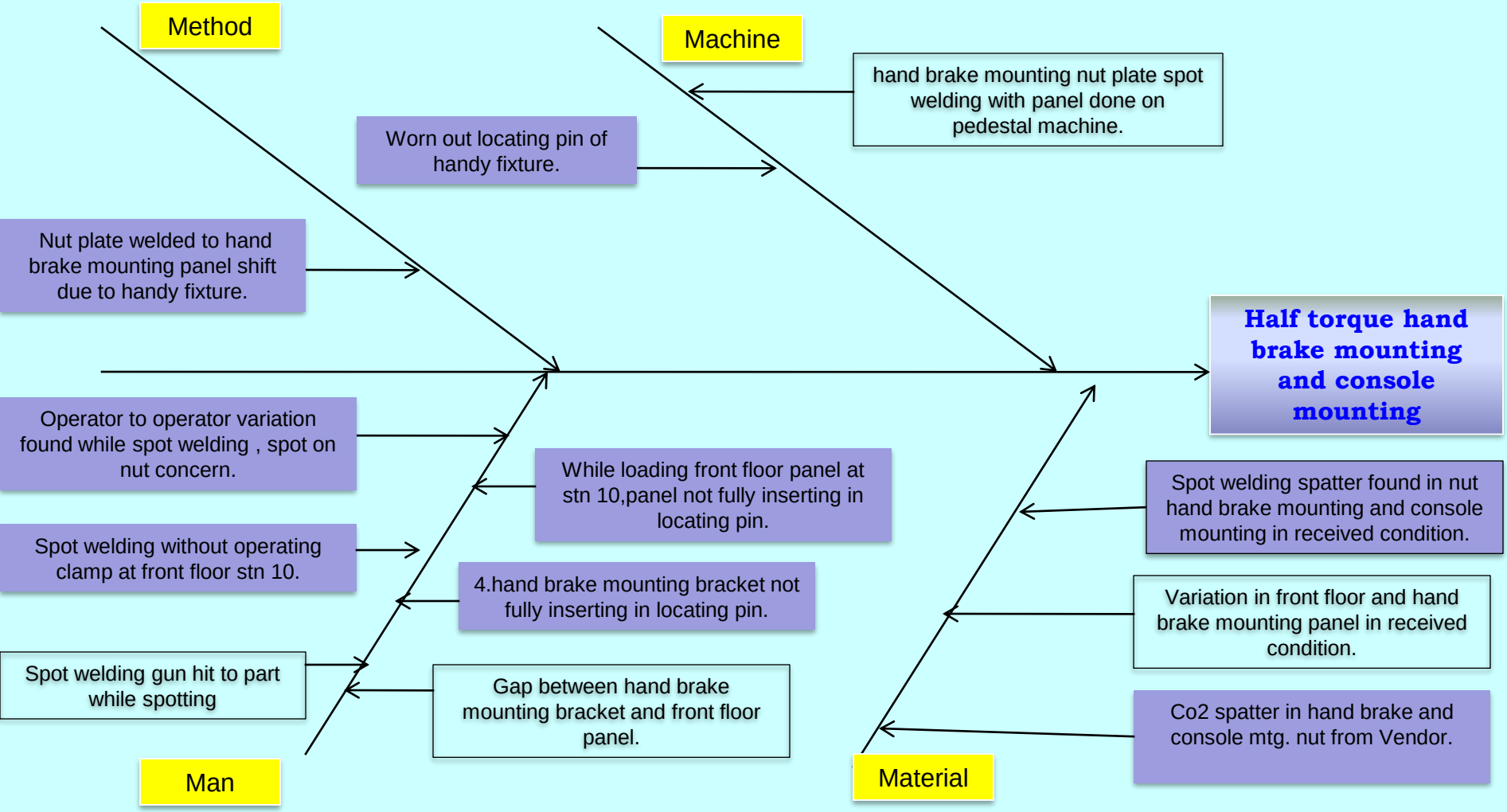
5. Analysis

Brainstorming by the CFT

Sr No	Observation	Possible cause
1.	Observations related to Handy fixture at MUSCO.	1.Nut plate welded to hand brake mounting panel shift due to handy fixture.
		2.Locating pin of handy fixture are in worn out condition.
2.	Observations related to process	3.While loading front floor panel at stn 10,panel not fully inserting in locating pin.
		4.hand brake mounting bracket not fully inserting in locating pin.
		5.Spot welding without operating clamp at front floor stn 10.
		5.At front floor stn 30 possibility of spot on nut hand brake mounting bracket.
3.	Spot welding	6.Operator to operator variation found while spot welding , spot on nut concern.
		7. At vendor hand brake mounting nut plate spot welding with panel done on pedestal machine.
4.	Observations related to material	8.Spot welding spatter found in nut hand brake mounting and console mounting in received condition.
		9. Variation in front floor and hand brake mounting panel in received condition.
		10. Co2 spatter in hand brake and console mtg. nut from Vendor.

5. Analysis

5.1 Identify probable causes:



TEST OF HYPOTHESIS

5.2 list of probable causes:

Sr. No	Probable Cause
1	Worn out locating pins of handy fixture at MUSCO.
2	Front floor panel loading on fixture at stn 10
3	Spot welding without operating clamp of hand brake and console mtg. at front floor stn 10.
4	Hand brake and console mounting bracket loading at stn 10.
5	Spot welding of hand brake mounting at front floor stn 30.
6	CO2 spatter in hand brake and console mtg. nut from Vendor.
7	Front floor panel and hand brake and console mtg panel hole CD as per drawing.
8	Nut plate welded to hand brake mounting panel shift due to handy fixture.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

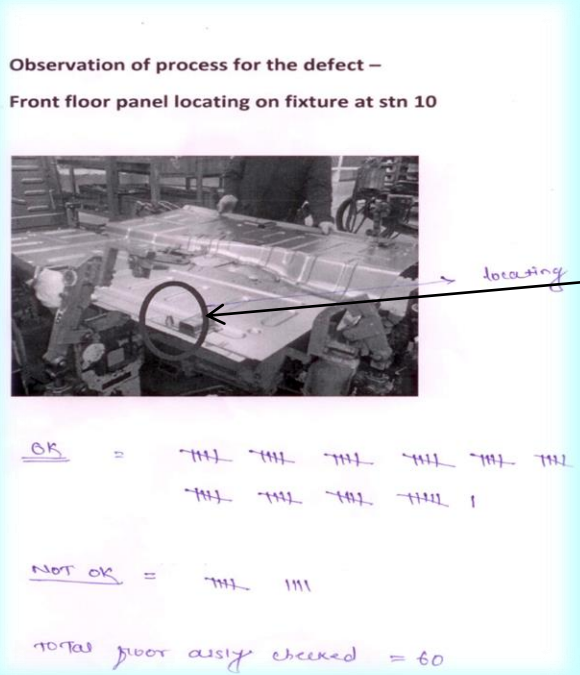
No	Probable Cause	Testing and Observations	Conclusion
1	Worn out locating pins of handy fixture at MUSCO.	Locating pins are worn out causing variation in nut plate welding.	**Valid**

Locating pin of hand brake and console mtg. bkt on dash panel found wear out & bend

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
2	Front floor panel loading on fixture at stn 10,	While loading front floor panel on fixture ,panel not fully inserting in locating pin.	**Valid**



Locating pin of front floor panel on fixture. found worn out (grove) causing variation in front floor panel mounting .

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
3	Hand brake and console mounting bracket loading and spot welding at stn 10.	Loading operation of hand brake and console mtg bracket at stn 10 followed by clamping operation , so there is no chance of position shift during spot welding.	**Invalid**

Clamp to hold hand brake mounting and console mtg brkt

TEST OF HYPOTHESIS

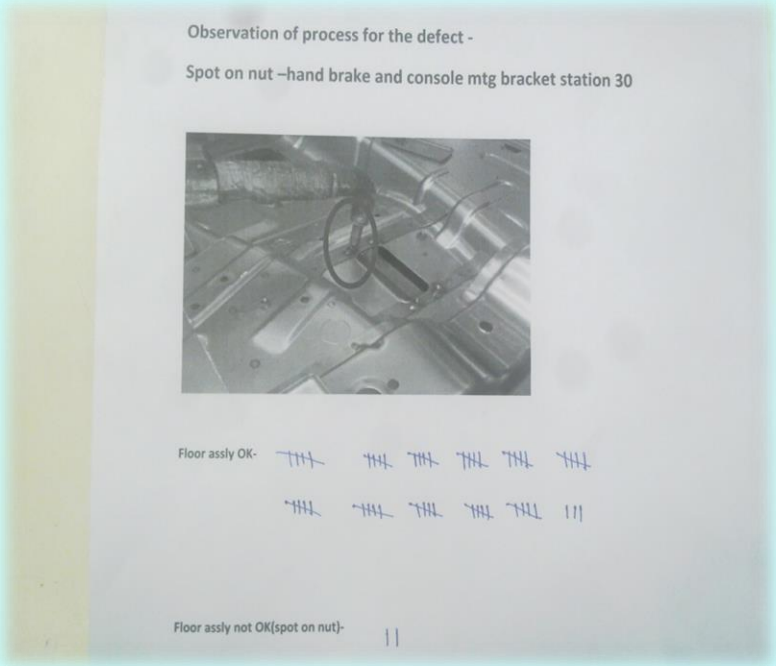
5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
4	Spot welding without operating clamp of hand brake and console mtg. at front floor stn 10. .	Interlocking of clamp with stud welding pokayoke	**Invalid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
5	Spot welding of hand brake mounting at front floor stn 30.	While Spot welding spot on front floor stn 30 spot on nut - hand brake mounting bracket.	**Valid**


Observation of process for the defect -

Spot on nut –hand brake and console mtg bracket station 30

Floor assly OK-

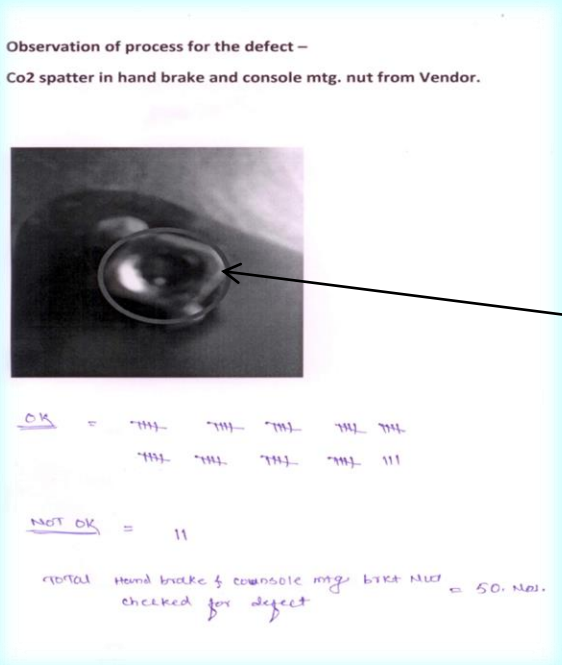
Floor assly not OK(spot on nut)-

Check sheet for spot on nut hand brake mounting and console mtg brkt

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
6	CO2 spatter in hand brake and console mtg. nut from Vendor.	50 Vendor parts checked on sampling basis found Co2 spatter in 2 parts	**Valid**



Observation of process for the defect –

Co2 spatter in hand brake and console mtg. nut from Vendor.

OK =

NOT OK = 11

Total Hand brake & console mtg. bkt Nut checked for defect = 50. Nos.

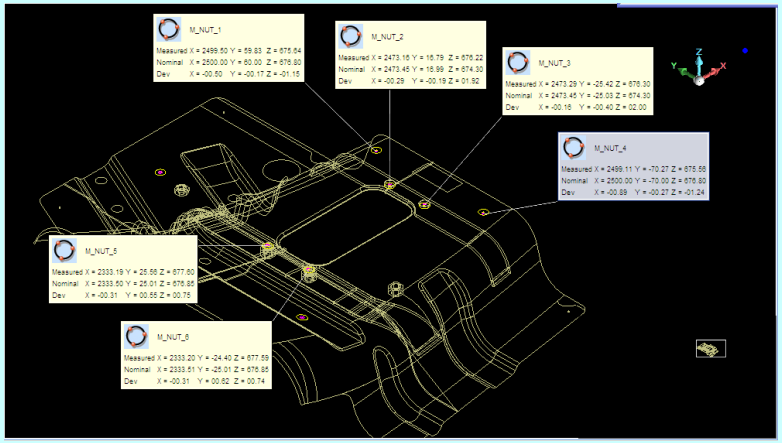
Spatters found in hand brake and console mtg. nut during Co2 welding at Vendor end.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
7	Front floor panel and hand brake and console mtg panel hole CD as per drawing.	Front floor panel and hand brake mtg brkt panel Checked on panel checker, no nonconformity Found.	**Invalid**

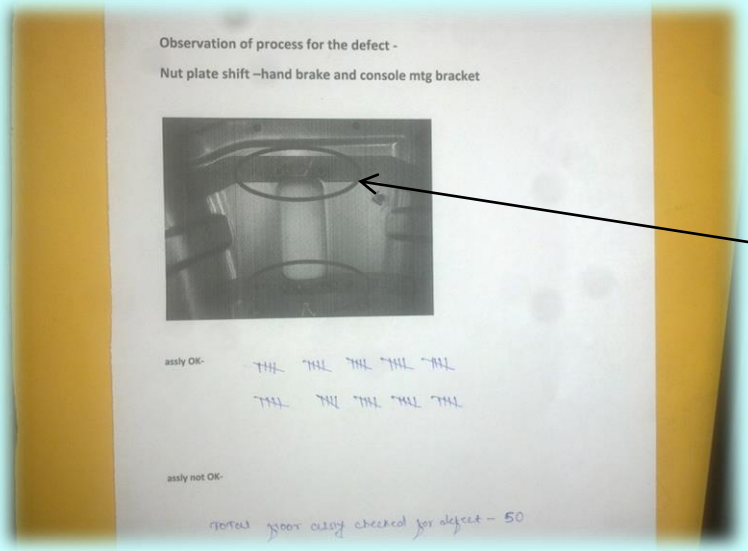
Front floor panel and hand brake mtg brkt panel checked on panel checker and CMM at supplier end.



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
8	Nut plate welded to hand brake mounting panel shift due to handy fixture.	50 Vendor parts checked on sampling basis no variation found.	**Invalid**



Nut plate welded to hand brake mounting panel
Checked for 50 parts at Vendor end.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Cause	Conclusion
1	Worn out locating pins of handy fixture at MUSCO.	Hypothesis Valid
2	Front floor panel loading on fixture at stn 10	Hypothesis Valid
3	Spot welding without operating clamp of hand brake and console mtg. at front floor stn 10.	Hypothesis Invalid
4	Hand brake and console mounting bracket loading at stn 10.	Hypothesis Invalid
5	Spot welding of hand brake mounting at front floor stn 30.	Hypothesis Valid
6	CO2 spatter in hand brake and console mtg. nut from Vendor.	Hypothesis Valid
7	Front floor panel and hand brake and console mtg panel hole CD as per drawing.	Hypothesis Invalid
8	Nut plate welded to hand brake mounting panel shift due to handy fixture.	Hypothesis Invalid

WHY - WHY Analysis

5.3 Identify the root causes:

Sr No	Probable cause	why	why	why	why	Action plan
1	Worn out locating pins of handy fixture at MUSCO.	Locating Pin undersize	Frequent wearing of the pin	Pin changing frequency not decided	Pin life not known	Locating pin change.
2	Front floor panel loading on fixture at stn 10	Grove to the locating pin	Frequent wearing of the pin	Pin changing frequency not decided	Pin life not known	Locating pin change.
3	Spot welding of hand brake mounting at front floor stn 30.	Difficult gun movement	Compact area of spotting	No guiding to the gun		Gun guide provided.
4	CO2 spatter in hand brake and console mtg. nut from Vendor.	Single locating pin is used for Co2 welding	No fool proof fixture available			1.Full proofing for fixture. 2.Parameter optimization

6. Implement the corrective measures

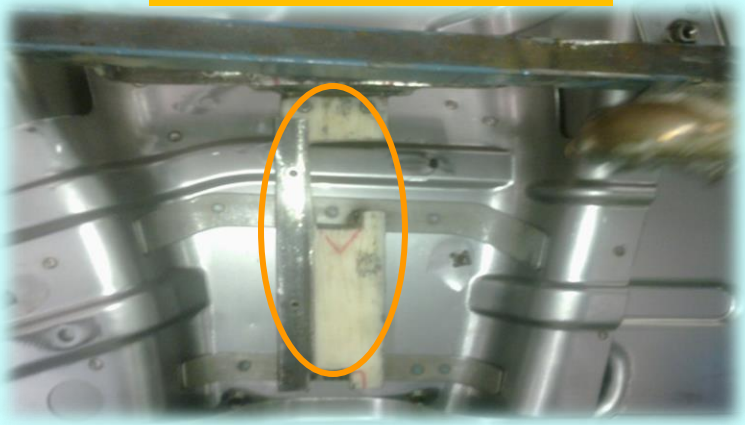
6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Spot taken on hand brake and console mtg. nut while spot welding at stn 30.	guard provided at Front floor stn.30 to avoid spot on nut during spot welding.	14.09.2012	U. kankal	Closed

Before

No guard to avoid spot on nut.

After



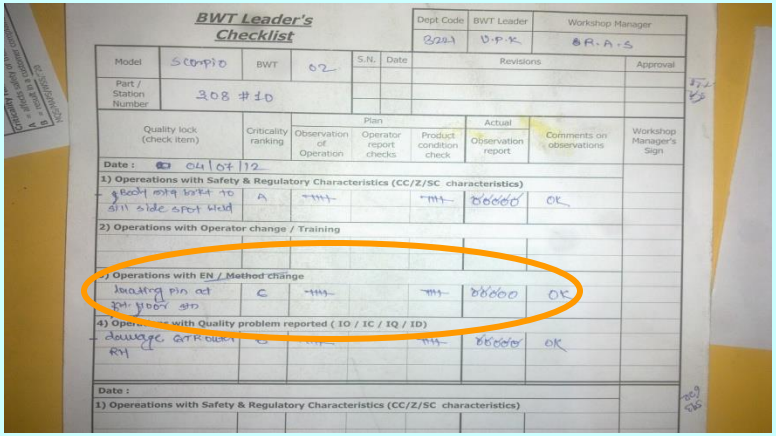
Guard to avoid spot on nut.

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Variation in front floor panel position during loading on fixture.	Locating pin worn out found (grove on locating pin), locating pin replaced. Check point added in Supervisory check sheet.	20.06.2012	U. kankal	Closed

New locating Pin provided.



BWT Leader's Checklist

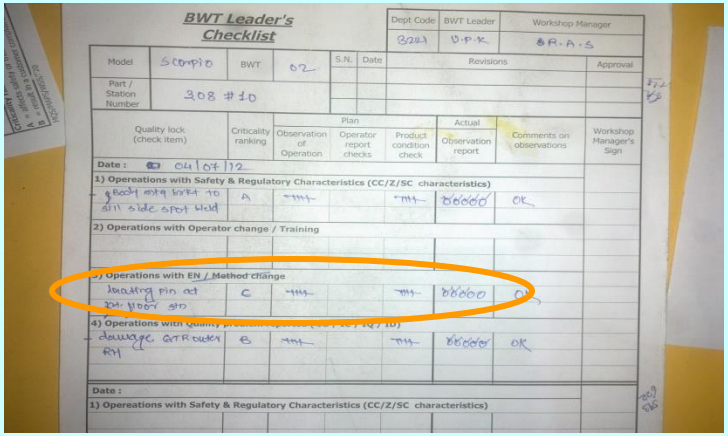
Supervisory check sheet.

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Worn out locating pins of handy fixture.	Locating pin worn out found, locating Pin replaced. Check point added in Supervisory check sheet.	24.06.2012	U. kankal	Closed

Locating Pin replaced.

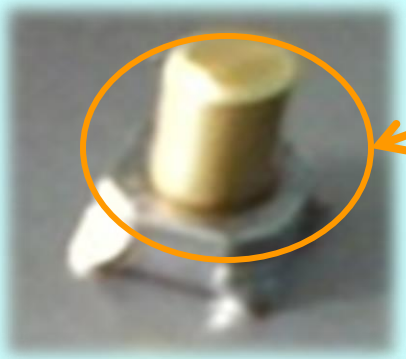


Supervisory check sheet.

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Co2 spatter inhand brake and console mtg. nut from Vendor.	ICA:- 100% Inspection & tapping at Vendor end. PCA:- 1.Providing new full proof fixture for Co2 welding 2.Co2 welding parameters are optimized	11.10.2012	D.Giri	Closed



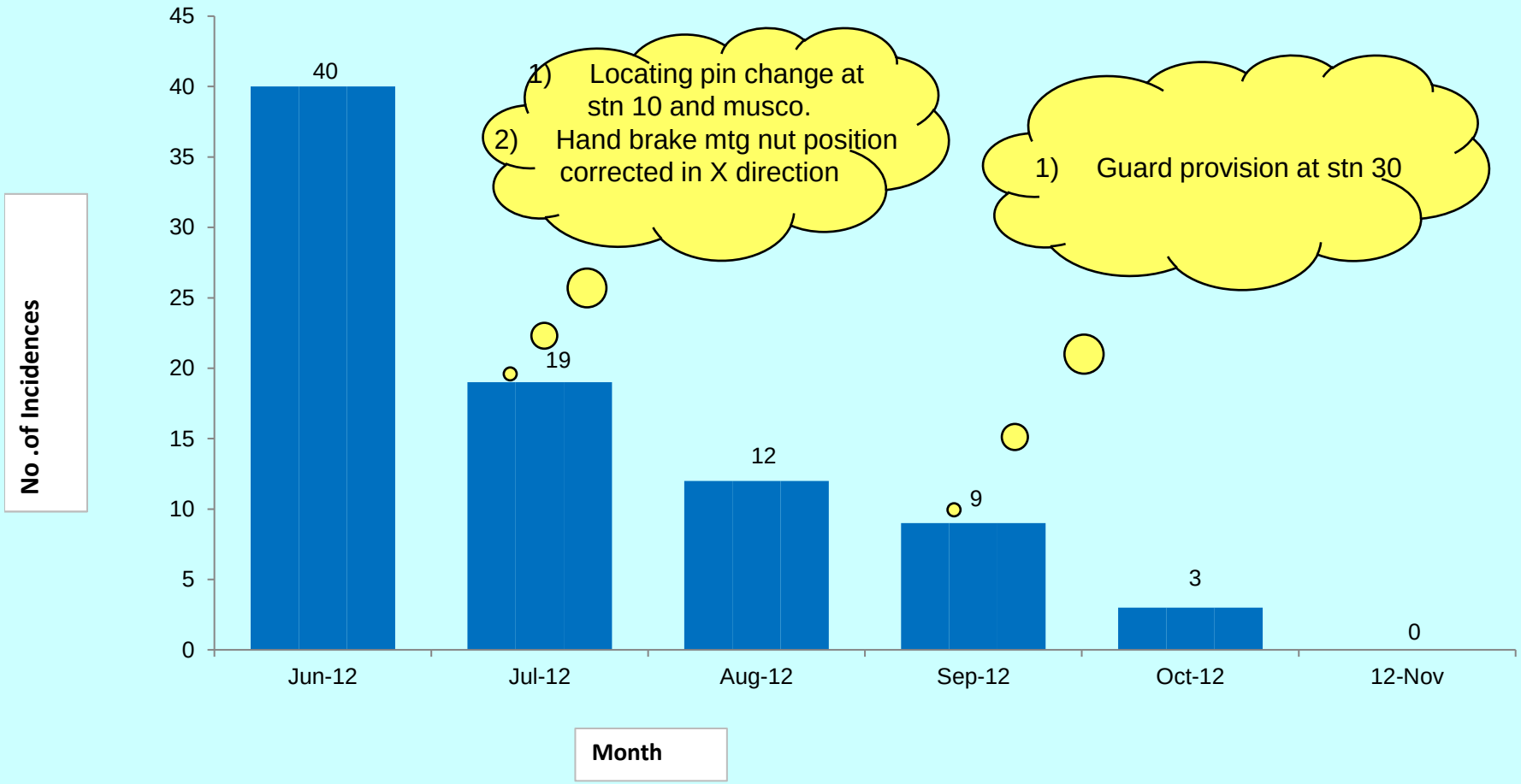
1.Brass Pin provided for masking in nut thred avoid to spatter during co2 tack welding operation.

2.Co2 welding parameters optimized

Before	After
C-120-130Amps	C- 100-120Amps
V-22-28 V	V-20-25 V
Gas flow-10-15Kg/cm sq	Gas flow-10-15Kg/cm sq

7. Check & confirm the results & effects

7.1 Verify Results:



• 7.2 Full Scale Implementation plan:

1. All decided actions implemented.

2. Effectiveness of actions has been audited on 29.10.2012 by MQA team

Check & confirm the results & effects (Cont……)

7.3 Tangible :

1) FRC improvement at TCF by 0.7%
2) COPQ reduction by 100116/annum.
3) High customer satisfaction through customer voice.

7.3 Intangible benefits:

1. Team Morale improved
2. High Customer satisfaction
3. Team Work Built up by supplier gamba visit.
4. Team Confidence Boost up.

8. Standardisation

8.1 Updation of documents:

1) Fixture History Card
2) PM Check Sheet (Locating Pin)

8.1 Updation of documents:

1) Fixture History Card

FIXTURE HISTORY CARD					
ZONE:	Z05				DEPT : MFG ENGG.-SCORPIO BODY
FIXTURE NO Z05 -F30			FREQUENCY : 6 Months and/or whenever changes done		
ASSY NAME Assy Front floor.					
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
29/3/02		F3-07-00	Speedo cable brkt added.	Unit added on fixture.	
20/04/2007			Wrong mtg. Floor to centre floor assy (spare body)	Pokayoke provided using 2 sensor circuit.	Heavy rework avoided.
10/12/2010			------	Center floor loc pin shifted in X-Direction by 1.00mm in +ve X-Direction..	As per CMM Report
15/8/11			Excess gap between Lh / Rh sill side to rear floor	Center floor to be shifted by 1.5 mm towards X(-) direction	
14/09/12			Spot on nut at hand brake mtg brt.	Guard Provided at front floor #30.	TCF Fitment

8.1 Updation of documents:

2) PM Check Sheet (Locating Pin)

M & M Ltd. Automotive Sector Nashik	QUALITY MANAGEMENT SYSTEM CHECK LIST TITLE : Preventive Maintenance of Fixtures-Framing	Dept : Mfg. Engg Page: 01 of 01
1.PURPOSE:	To Ensure the Preventive Maintenance of Fixtures Carried-out Correctly.	
2.SCOPE :	All Framing line Fixtures in Scorpio Body shop & SCDC	

Assy Name: **Stage:**
Fixture Name:
Fixture No. **Date :-**

Sr. No.	Check Points	Work Instruction	Status	Remark
1	Fixture cleaning.	Clean thoroughly all areas of Fixture (Resting points/location pins /Guides/clamps/base frame/valves etc.		
2	Production Feed back	Check for any corrective actions reqd. as per the production feed back.		
3	Fixture structure	Check for Bend,Distort, Breakage etc.		
4	Fastener checking.	Ensure the tightness of fastners		
		Marking on the Fasteners.		
		Check for spring washer.		
5	Dowel checking	Check for proper fitment of dowels		
		Ensure fitment of circlip on the dowels.		
		Check fitment of cover plate on Dowels & Allen screws.		
6	Locator/Resting block/Spacers/Nylon strips.	Check locator fitment.		
		Check for any breakage/wearout		
		Check for proper fitment of pins.		
		Check for any breakage/wearout of pin		
		Check for magnet breakage & its magnetic power.		
7	Pneumatic cylinders	Check for fitment of cylinder bolts/bracket		
		Check hinge pin of cylinder for washer,cotter pin.		
		Check for piping leakage/fitment.		
		Check for up / down movement		
8	Clamping block	Check hinge pins of clamps for washer,cotter pin.		
		Check for proper clamping of clamping blocks on resting blocks		
		Check guide blocks		
		Check kinifoam applied on resting myler		
9	LM Guides	Check fitment of LM guides/dowels.		
		Check position/fitment of stopper blocks for LM guide block.		
10	Limit switches/Solenoid valves/Sensors check	Check for proper mounting/damage		
		Check for wiring/pipe connection.		
11	Greasing /Oiling	Do Greasing/oiling at all moving & rotating areas.		
12	Lifters	Check cylinder mtg/Guide bush /Guide bar		
		Check structure for bending/bolting.		
		Check rollers condition,levelling.		
13	POKA-YOKE	Check for POKA -YOKE points.		

Note:-(Please write your observation / comment if any other than check points or work done on Fixture)

Operator Name :- **Operator Signature :-**

Sbody/ME/F/03 , Rev 2.10, Eff.28//06/2011

8.2 Training:

1. OPL given to all stage operators.
2. Operator were trained to handle the gun at front floor stn 30 to prevent spot damage and gun hitting to nut.

8.3 Additional Audit / Check

1. Supplier Audit
2. PM Check Sheet (Locating Pin)

9. Summarise & Plan future actions

9.1 Conclusion:

An innovative approach and team work by doing GEMBA study can lead to elimination of such chronic concerns.

9.2 Horizontal Deployment:

Project initiated jointly with A. Shaikh for Side Assly LH/RH Jump Seat Mounting half torque issue

9.3 Lessons Learnt:

Things Gone Wrong :

Hand brake and console mtg brkt and floor panel mismatch at front floor stn 10.

Things Gone Right :

We study the process implemented the gun guide without much disturbance to gun movement and operator fatigue.

THANK YOU